PUSHFORWARD PRODUCT FORMULA FOR FLAT REPRESENTABLE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4737-4747 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 4749-4789, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 27 of queue_stacks-stacks-more-morphisms.json (pb=1983, stmt 4737-4747, proof 4749-4789).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FLQ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4789–4789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06UM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4789–4789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0771: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4789–4789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GQM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4789–4789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GQS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 4789–4789. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $a : \mathcal{Y} \to \mathcal{X}$ and $b : \mathcal{Z} \to \mathcal{X}$ be representable by schemes, quasi-compact, quasi-separated, and flat. Then $a_{QCoh,*}\mathcal{O}_\mathcal{Y} \otimes_{\mathcal{O}_\mathcal{X}} b_{QCoh,*}\mathcal{O}_\mathcal{Z} = f_{QCoh,*}\mathcal{O}_{\mathcal{Y}\times_\mathcal{X}\mathcal{Z}}$ where $f : \mathcal{Y} \times_\mathcal{X} \mathcal{Z} \to \mathcal{X}$ is the obvious morphism.*

Proof. We abbreviate $\mathcal{P} = \mathcal{Y} \times_\mathcal{X} \mathcal{Z}$. Since $a \circ \mathrm{pr}_1 = f$ and $b \circ \mathrm{pr}_2 = f$ we obtain maps $a_*\mathcal{O}_\mathcal{Y} \to f_*\mathcal{O}_\mathcal{P}$ and $b_*\mathcal{O}_\mathcal{Z} \to f_*\mathcal{O}_\mathcal{P}$ (using relative pullback maps, see Sites, Section 06UM). Hence we obtain a relative cup product

$$\mu : a_*\mathcal{O}_\mathcal{Y} \otimes_{\mathcal{O}_\mathcal{X}} b_*\mathcal{O}_\mathcal{Z} \longrightarrow f_*\mathcal{O}_{\mathcal{Y}\times_\mathcal{X}\mathcal{Z}}$$

Applying Q and its compatibility with tensor products (Cohomology of Stacks, Remark 0GQM) we obtain an arrow $Q(\mu) : a_{QCoh,*}\mathcal{O}_\mathcal{Y} \otimes_{\mathcal{O}_\mathcal{X}} b_{QCoh,*}\mathcal{O}_\mathcal{Z} \to f_{QCoh,*}\mathcal{O}_{\mathcal{Y}\times_\mathcal{X}\mathcal{Z}}$ in $QCoh(\mathcal{O}_\mathcal{X})$. Next, choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. It suffices to prove the restriction of $Q(\mu)$ to $U_{étale}$ is an isomorphism, see Cohomology of Stacks, Section 0GQS. In turn, by the material in the same section, it suffices to prove the restriction of μ to $U_{étale}$ is an isomorphism (this uses that the source and target of μ are locally quasi-coherent modules with the base change property). Moreover, we may compute pushforwards in the étale topology, see Cohomology of Stacks, Proposition 0771. Then finally, we see that $a_*\mathcal{O}_\mathcal{Y}|_{U_{étale}} = (V \to U)_{small,*}\mathcal{O}_V$ where $V = U \times_\mathcal{X} \mathcal{Y}$. Similarly for b_* and f_*. Thus the result follows from the Künneth formula for flat, quasi-compact, quasi-separated morphisms of schemes, see Derived Categories of Schemes, Lemma 0FLQ. □